



CONCRETE COMPRESSIVE STRENGTH PREDICTION USING MACHINE LEARNING

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1. Problem Definition

Concrete is one of the most widely used construction materials. The compressive strength of concrete depends on various factors such as cement content, water content, aggregates, and curing age.

The objective of this project is to develop a Machine Learning model capable of predicting the compressive strength of concrete based on its constituent materials and age. Accurate prediction of concrete strength can help engineers optimize concrete mixtures and improve construction quality.

2. Dataset Description

The Concrete Compressive Strength dataset was used for this project. The dataset contains concrete mixture information along with the corresponding compressive strength values.

Dataset Source:

<https://www.kaggle.com/datasets/niteshyadav3103/concrete-compressive-strength>

Number of Records

Number of Records :1005

Features Used:

Feature	Description
Cement	Cement content in kg/m^3
Blast Furnace Slag	Slag content in kg/m^3
Fly Ash	Fly ash content in kg/m^3
Water	Water content in kg/m^3
Superplasticizer	Chemical admixture content
Coarse Aggregate	Coarse aggregate content
Fine Aggregate	Fine aggregate content
Age	Age of concrete in days

Target Variable:

Concrete Compressive Strength (MPa)

Data Preprocessing

Several preprocessing steps were performed before training the machine learning model.

Step 1: Data Loading

The dataset was loaded into Python using the Pandas library.

Step 2: Duplicate Record Removal

Duplicate data is not observed in the dataset as the `data.isnull().sum()` is showing zero.

Step 3: Feature and Target Separation

The dataset was divided into:

- Input Features (X)
- Target Variable (y)

Step 4: Feature Scaling

Since the feature values had different ranges. `StandardScaler` was applied to normalize the data.

Step 5: Train-Test Split

The dataset was split into:

- Training Data: 80%
- Testing Data: 20%

This ensured unbiased evaluation of the model.

```
'Concrete compressive strength'],
dtype='str')
Cement Blast Furnace Slag Fly Ash Water Superplasticizer Coarse Aggregate Fine Aggregate Age (day) Concrete compressive strength
0 540.0 0.0 0.0 162.0 2.5 1040.0 676.0 28 79.986111
1 540.0 0.0 0.0 162.0 2.5 1055.0 676.0 28 61.887366
2 332.5 142.5 0.0 228.0 0.0 932.0 594.0 270 40.269535
3 332.5 142.5 0.0 228.0 0.0 932.0 594.0 365 41.052780
4 198.6 132.4 0.0 192.0 0.0 978.4 825.5 360 44.296075
(1005, 9)
Cement 0
Blast Furnace Slag 0
Fly Ash 0
Water 0
Superplasticizer 0
Coarse Aggregate 0
Fine Aggregate 0
Age (day) 0
Concrete compressive strength 0
dtype: int64
Training data shape: (804, 8)
Testing data shape: (201, 8)
PS C:\Users\Aleni\OneDrive\Pictures\Documents\AI_ML_INTERNSHIP> 
```

4. Model Building

Selected Model

Random Forest Regressor

Reason for Selection

Random Forest Regressor was selected because:

- It can model complex non-linear relationships.
- It reduces overfitting through ensemble learning.
- It generally provides higher prediction accuracy than simple linear models.
- It performs well on regression tasks involving multiple numerical features.

Model Training

The model was trained using the training dataset and then used to predict concrete compressive strength values on the test dataset.

MODEL COMPARISON

Linear Regression
R² Score : 0.5801
MAE : 8.90
RMSE : 11.19

Best KNN (k=3)
R² Score : 0.7467
MAE : 6.43
RMSE : 8.69

Best Random Forest (n_estimators=28)
R² Score : 0.9174
MAE : 3.42
RMSE : 4.96

Best Overall Model: Random Forest

5. Model Evaluation

The performance of the model was evaluated using standard regression metrics.

Metric	Value
Mean Absolute Error (MAE)	3.12
Root Mean Squared Error (RMSE)	4.96
R ² Score	0.9174

Interpretation

The obtained R² score indicates how well the model explains the variation in concrete compressive strength.

A lower MAE and RMSE indicate that the predicted values are close to the actual values.

The evaluation results demonstrate that the Random Forest Regressor is capable of predicting concrete strength with good accuracy.

6. Conclusion

In this project, a Random Forest Regressor was developed to predict concrete compressive strength using material composition and curing age.

The model successfully learned the relationship between input variables and concrete strength and achieved satisfactory performance on the test dataset.

The results indicate that machine learning techniques can be effectively used for predicting concrete strength and assisting engineers in material design and decision-making.

7. Limitations

1. The model performance depends on the quality of the dataset.
2. Only the available features were considered.
3. Additional environmental factors may influence concrete strength but were not included.

8. Reflection

This project helped me understand the complete machine learning workflow, including data preprocessing, feature selection, model training, and performance evaluation. I

learned how to handle datasets, remove duplicate records, perform feature scaling, and split data into training and testing sets.

One of the challenges faced during implementation was understanding the dataset structure and selecting an appropriate machine learning algorithm. Through experimentation, I learned that Random Forest Regressor is effective for predicting concrete compressive strength because it can capture complex relationships between variables.

I also gained practical experience with regression evaluation metrics such as MAE, MSE, RMSE, and R^2 Score. These metrics helped me understand how well the model performs and how prediction errors can be measured.

Overall, this project strengthened my understanding of the machine learning pipeline and provided valuable hands-on experience in applying machine learning techniques to an engineering-related problem

PYTHON CODE:

```
1 from sklearn.linear_model import LinearRegression
2 from sklearn.neighbors import KNeighborsRegressor
3 from sklearn.ensemble import RandomForestRegressor
4 from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error
5 from sklearn.model_selection import train_test_split
6 from sklearn.preprocessing import StandardScaler, LabelEncoder
7 import pandas as pd
8 import joblib as jb
9 import numpy as np
10
11 data = pd.read_csv("Mini_Project_1/Concrete_data.csv")
12 data = data.drop_duplicates()
13 #print(data.columns)
14 #data preprocessing
15 x = data.drop('Concrete compressive strength', axis=1)
16 y = data["Concrete compressive strength"]
17 #print(data.head())
18 #print(data.shape)
19 #print(data.isnull().sum())
20 x_train, x_test, y_train, y_test = train_test_split(
21     x, y, test_size=0.2, random_state=42)
22
23 scaler = StandardScaler()
24 x_train_scaled = scaler.fit_transform(x_train)
25 x_test_scaled = scaler.transform(x_test)
26 '''
27 print("Training data shape:", x_train.shape)
28 print("Testing data shape:", x_test.shape)'''
29
30 linear = LinearRegression()
31 linear.fit(x_train_scaled, y_train)
32
33 linear_pred = linear.predict(x_test_scaled)
34 linear_r2 = r2_score(y_test, linear_pred)
35 linear_mae = mean_absolute_error(y_test, linear_pred)
36 linear_rmse = np.sqrt(mean_squared_error(y_test, linear_pred))
37
38 best_knn = None
39 best_knn_r2 = float("-inf")
40 best_k = 0
41
42 for i in range(1, 21, 1):
43     knn = KNeighborsRegressor(n_neighbors=i)
44     knn.fit(x_train_scaled, y_train)
45     pred = knn.predict(x_test_scaled)
```

```

Mini_Project_1 > Concre.py > ...
75
76 print("\nMODEL COMPARISON")
77 print("-" * 40)
78 print("\nLinear Regression")
79 print(f"R² Score : {linear_r2:.4f}")
80 print(f"MAE      : {linear_mae:.2f}")
81 print(f"RMSE     : {linear_rmse:.2f}")
82 print('-' * 40)
83 print(f"\nBest KNN (k={best_k})")
84 print(f"R² Score : {best_knn_r2:.4f}")
85 print(f"MAE      : {knn_mae:.2f}")
86 print(f"RMSE     : {knn_rmse:.2f}")
87 print('-' * 40)
88 print(f"\nBest Random Forest (n_estimators={best_estimators})")
89 print(f"R² Score : {best_rf_r2:.4f}")
90 print(f"MAE      : {rf_mae:.2f}")
91 print(f"RMSE     : {rf_rmse:.2f}")
92 print('-' * 40)
93 model = {
94     "Linear Regression": (linear, linear_r2),
95     "KNN": (best_knn, best_knn_r2),
96     "Random Forest": (best_rf, best_rf_r2)}
97
98 best_model_name = max(model, key=lambda x: model[x][1])
99 best_model = model[best_model_name][0]
100 print(f"\nBest Overall Model: {best_model_name}")
101 print('-' * 40)
102 job.dump(best_model, "best_concrete_model.pkl")
103 job.dump(scaler, "concrete_scaler.pkl")
104

```

OUTPUT:

```

PS C:\Users\Aleni\OneDrive\Pictures\Documents\AI_ML_INTERNSHIP> & C:\Users\Aleni\AppData\Local\Programs\Python\Python312\python.exe c:/Users/Aleni/OneD
rive/Pictures/Documents/AI_ML_INTERNSHIP/Mini_Project_1/concrete_input.py
Enter Cement: 540
Enter Blast Furnace Slag: 0
Enter Fly Ash: 0
Enter Water: 162
Enter Superplasticizer: 2.5
Enter Coarse Aggregate: 1040
Enter Fine Aggregate: 676
Enter Age (days): 28
*****

Predicted Concrete Strength: 72.86 MPa
*****
PS C:\Users\Aleni\OneDrive\Pictures\Documents\AI_ML_INTERNSHIP>

```